

Harsh Bali

Backend Engineer — Real-Time & Distributed Systems

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Summary

Backend engineer specialising in real-time, distributed systems. Built event-driven architectures with Redis, Kafka, and WebSockets — achieving sub-second P95 latency, 100 real-time updates/sec, and zero message loss across production systems supporting 1k+ concurrent users.

Experience

DocPharma

Jul 2026 – Present

Software Development Engineer – 1

- Spearheading high-concurrency backend services, WMS integrations, and query performance engineering for large-scale pharmaceutical & supply-chain workflows.
- Optimised slow database queries from 10s to under 0.1s (99% latency reduction) via targeted indexing; implemented PostgreSQL GIN indexes for fast pattern-matching and full-text search across large catalogues.
- Integrated core WMS services for end-to-end order fulfilment; enhanced distance calculation logic by offloading to dedicated distance APIs, reducing backend load and improving accuracy.

Idea Usher

May 2024 – Jun 2026

Backend Engineer

- Scaled event-driven backend (Node.js, Redis, WebSockets) to 1k+ concurrent users while leading a monolith-to-microservices migration; re-architected Redis pub/sub adapter with connection pooling & backpressure control, cutting P95 latency from ~15s to under 1s.
- Built async BullMQ + Redis job pipelines with rate limiting, delayed jobs, and fault-tolerant retries; WebSocket layer sustained 100 real-time price updates/sec with zero dropped events.
- Delivered a FIFO order-matching engine with live P&L streaming (WebSockets + Redis Streams) for 1k+ concurrent traders at sub-100ms latency; refactored a real-time poker platform from monolith to modular architecture in 1 week, eliminating critical race conditions.

Projects

Decentralized Social Media Platform | *NestJS, gRPC, Redis, Docker & K8s, AWS*

- Engineered asynchronous fan-out-on-write architecture for real-time feed distribution alongside a media ETL pipeline for video transcoding and storage.
- Containerised microservices backend on Docker & Kubernetes; gRPC for sub-10ms internal communication with distributed rate limiting and Bloom filters.

Skills

Languages: TypeScript, JavaScript, Java, SQL, C++ **Backend:** Node.js, NestJS, Express, REST, gRPC, WebSockets, Microservices

Systems: Redis (pub/sub, streams, caching), BullMQ, Kafka, Event-Driven Architecture **Databases:** PostgreSQL, MySQL, MongoDB

DevOps: Docker, Kubernetes, AWS, Azure, CI/CD **AI:** RAG, OpenAI APIs, Vector Databases, ChromaDB

Achievements

3rd place — Xpression Hackathon (state-level) for building an AI-powered application.

Guest speaker on backend architecture at CSJM University; conducted DSA training workshops for undergraduates on problem-solving patterns.

Education

CSJM University

2020 – 2024

Bachelor of Computer Applications (BCA)

CGPA: 8.92 / 10